# RAG Evaluation Report


# RAG Evaluation Report


**Run:** `run1` · **Generated:** 2026-06-15 13:35 · **Status:** complete, all services stopped (no ongoing cost)

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## Executive summary

A live, end-to-end evaluation of the document-RAG pipeline over **3 operation manuals** (1,021 indexed chunks), answering a **27-question gold set** with physical-page citations. Visual table/figure summaries were generated by the **fine-tuned Qwen3-VL-4B** model, self-hosted on a Modal A10G GPU. Answers were scored by retrieval/citation/answer/ops metrics plus a **gpt-4o LLM judge**.

**Verdict:** a **trustworthy retrieval system** — it reliably finds the right page (Recall@5 **96%**), stays grounded (faithfulness **4.89/5**), and refuses cleanly when the answer isn't in the corpus (no-answer accuracy **100%**). The weak points are **citation precision** and **dense academic-paper content** (the GPT paper), where the only 2 hallucinations and 3 fact-misses occurred.

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## Setup

| | |
|---|---|
| Corpus | UR5_handbook.pdf (294 chunks) · teslaOwnManual.pdf (572) · gpt.pdf (155) = **1,021** |
| Gold set | 27 Q — 18 factual / 4 procedure / 2 suggestion / 3 no-answer |
| Retrieval | dense (`text-embedding-3-small`) + query-enhancement (HyDE/decomposition) |

| | |
|---|---|
| Synthesis | `gpt-4.1-mini`, multi-agent (table/figure specialists) |
| VLM summaries | **fine-tuned Qwen3-VL-4B** (`azhuang3/qwen3_vlm_task`) on Modal A10G — 644 regions enriched |
| Judge | `gpt-4o` (stronger than generator, reduces self-bias) |
| Defaults | hybrid / reranker / compression / faithfulness-check all **OFF** |

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## 1 · Retrieval quality

| Metric | Value | |
|---|---|---|
| MRR | **0.77** | first relevant hit near rank 1–2 |
| Recall@1 | 54% | right page is #1 half the time |
| Recall@3 | 83% | |
| **Recall@5** | **96%** | right page almost always in top-5 |
| nDCG@5 (graded) | 0.79 | good ranking of graded relevance |

✅ **No question had a complete retrieval miss** (every gold page appeared in top-5 for at least one citation). The gap between Recall@1 (54%) and Recall@5 (96%) means the right page is usually retrieved but not always ranked first — a **reranker** would lift R@1.

## 2 · Citation quality *(top-2 cited sources)*

| | Page-tolerant (±1) | Page-exact |
|---|---|---|
| Precision | 62% | 46% |
| Recall | 67% | 65% |

⚠️ **This is the weakest area.** Answers are correct but the *cited* page is often a neighbor of the true page. For a handbook assistant where users click to verify, this matters most.

## 3 · Answer quality

**Deterministic checks** | Metric | Value | |—|—| | Must-include pass rate | 89% (24/27) | | Must-not-include violations | 0% | | **No-answer accuracy** | **100%** (3/3 unanswerable Qs correctly refused) | | False-refusal rate | 4% (1/24) |

**LLM judge (gpt-4o, 1–5)** | Metric | Value | |—|—| | Answer correctness | **4.89** | | Faithfulness / groundedness | **4.89** | | Completeness | 4.85 | | Hallucination rate | **7%** (2/27) |

## 4 · Suggestion quality *(engineering recommendations)*

- Evidence-supported (cites a manual page): 50%
- Judge — actionable **5.00/5**, grounded in cited evidence 100%

## 5 · Operational

| Metric | Value |
|---|---|
| Latency mean | **10.7s** |
| Latency p50 / p95 / max | 9.4s / 17.3s / 32.5s |
| Est. cost (approx) | $0.26 / 27 Q (~$0.009/query) |

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## Where it failed (all on the dense GPT academic paper)

| Question | Issue |
|---|---|
| `gpt1_architecture` — params of GPT-1 | missed “117M”; judge corr 3/5, **hallucination** |
| `gpt3_corpus` — GPT-3 training corpus | judge corr 4/5, **hallucination** |
| `ur5_temperature` — operating temp range | missed exact “0-50” string (likely phrasing/units) |
| `tesla_tire_mileage_suggestion` | missed “speed limit” anchor (still a reasonable suggestion) |

**Pattern:** the manuals (UR5, Tesla) are clean; the **dense 2-column academic PDF is the hard case** — OCR + retrieval struggle with packed tables/equations, which is exactly where the few errors land.

## Honest caveats

1. **27 questions is directional** (±10–15% per metric). Expand to 50–100 for stable deltas.
2. **Judge scores (~4.9/5) run high** — LLM judges are generous and the gold set was model-authored. Human-audit ~10 questions to calibrate before trusting absolute values.
3. **Qwen VLM value is unmeasured** — no question isolates table/figure reasoning, and no VLM-on/off ablation was run. The pipeline works (644 regions enriched, verified output) but its *contribution* is unquantified.
4. **Cost is a rough char/4 estimate** (the OpenAI client doesn't expose token usage).
5. **Defaults only** — hybrid/rerank/compression OFF.

## Recommended next steps

1. `ablation_eval` — quantify what query-enhancement buys (vs. the 10.7s latency) and what's safe to cut.
2. **Add a reranker** — directly targets the two weak spots (citation precision + Recall@1).
3. **Table/figure question set + VLM on/off ablation** — finally measure the Qwen pipeline's value.
4. **Grow the gold set to 50–100** + human-audit ~10 to calibrate the judge.

## Notes on metric integrity

Two eval-framework bugs were found and fixed *during* this run (committed): - nDCG could exceed 1.0 when multiple chunks hit one gold page → now each gold credited once. - Refusal detector missed "do not provide / not stated" → no-answer accuracy corrected from 0% to the true 100%.

## Reproduce

```
.venv/bin/python -m eval.runners.retrieval_eval --gold eval/datasets/manual_questions.yaml \
    --predictions eval/reports/run1/predictions.json   # offline, free
```

Artifacts: `eval/reports/run1/` — REPORT.md, {retrieval,citation,answer,ops}_metrics.json, llm_judge.json, predictions.json.